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Light-emitting device and method for manufacturing light-emitting device

Abstract

A light-emitting device includes a plurality of plate-shaped base materials, an insulating coating film, and at least one light-emitting element. The base materials are arranged side by side to be mutually spaced. The insulating coating film is formed to cover an upper surface and a side surface of each of the plurality of base materials. The coating film is provided with an opening portion that exposes one region of the upper surface of one base material and includes a binding portion mutually binding the plurality of base materials. The at least one light-emitting element is placed on the one region. The coating film includes a thin film portion in which the coating film is formed in a thin film to surround an outer peripheral end of the opening portion.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a light-emitting device and a method for manufacturing a light-emitting device.
BACKGROUND ART
(2) There has been known a light-emitting device using a light-emitting element, such as a Light Emitting Diode (LED) or a Laser Diode (LD) as a light source.
(3) For example, WO-A1-2008/081794 discloses a light-emitting device including a light-emitting element, first and second lead frames, and first and second resin formed bodies. WO-A1-2008/081794 discloses that the light-emitting element is arranged in a predetermined arrangement region on the first lead frame in a casing formed as a molding.
DISCLOSURE OF THE INVENTION
Problems to be Solved by the Invention
(4) However, when an area of the arrangement region of the light-emitting element formed on the lead frame is significantly different from an area of the light-emitting element arranged in the region like the light-emitting device disclosed in WO-A1-2008/081794, the light-emitting element cannot be self-aligned, and the light-emitting element possibly deviates from a desired position. That is, the deviation of the arrangement position of the light-emitting element possibly causes a misalignment of an optical axis of a light emitted from the light-emitting device.
(5) The present invention has been made in consideration of the above-described points, and an object of which is to provide a light-emitting device and a method for manufacturing a light-emitting device capable of suppressing a positional deviation in bonding a light-emitting element on an arrangement region and stably bonding the light-emitting element to a predetermined position.
Solutions to the Problems
(6) A light-emitting device according to the present invention includes a plurality of plate-shaped base materials, an insulating coating film, a placing pad, and at least one light-emitting element. The plurality of plate-shaped base materials are arranged side by side to be mutually spaced. The plurality of base materials are each made of a metal. The insulating coating film is formed so as to cover an upper surface and a side surface of each of the plurality of base materials. The coating

film is provided with an opening portion that exposes one region of the upper surface of one base material among the plurality of base materials. The coating film includes a binding portion mutually binding the plurality of base materials. The placing pad is disposed so as to cover the one region of the upper surface of the one base material. The placing pad is made of a metal. The at least one light-emitting element is placed on the placing pad via a bonding member. The coating film includes a thin film portion in which the coating film is formed in a thin film so as to surround an outer peripheral end of the opening portion.

(7) A method for manufacturing a light-emitting device according to the present invention includes: a substrate forming step of funning a plurality of plate-shaped base materials that are arranged side by side to be mutually spaced and are each made of a metal, and forming an insulating coating film formed so as to cover an upper surface and a side surface of each of the plurality of base materials, the coating film being provided with an opening portion that exposes one region of the upper surface of one base material among the plurality of base materials, the coating film including a binding portion mutually binding the plurality of base materials; a plating step of providing a placing pad made of a metal so as to cover the one region of the upper surface of the one base material; and an element bonding step of placing at least one light-emitting element on the placing pad via a bonding member. In the substrata forming step, a thin film portion in which the coating film is formed in a thin film so as to surround an outer peripheral end of the opening portion is formed in the coating film.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a top view of a light-emitting device according to an embodiment of the present invention;

(2) FIG. **2** is a cross-sectional view of the light-emitting device taken along a line A-A of FIG. **1**;

(3) FIG. **3** is an enlarged view of a peripheral portion PA of an element placing portion of FIG. **2**;

(4) FIG. **4** is a drawing illustrating a flow chart of manufacturing the light-emitting device according to the embodiment of the present invention;

(5) FIG. **5** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(6) FIG. **6** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(7) FIG. **7** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(8) FIG. **8** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(9) FIG. **9** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(10) FIG. **10** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(11) FIG. **11** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(12) FIG. **12** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(13) FIG. **13** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention;

(14) FIG. **14** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention; and

(15) FIG. **15** is a cross-sectional view of the light-emitting device during the manufacture according to the embodiment of the present invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

(16) An embodiment of the present invention will be described in detail below. Note that the same reference numerals are given to substantially the same or equivalent parts in the following description and the accompanying drawings.

(17) FIG. **1** is a schematic top view of a light-emitting device **100**. FIG. **2** is a cross-sectional view of the light-emitting device **100** taken along a line A-A of FIG. **1**. In the following description, a description of “material 1/material 2” means a laminated structure in which a material 2 is laminated on a material 1. A description of “material 1-material 2” means an alloy of materials 1 and 2, and a description of “material 1-material 2-material 3” means an alloy of materials 1 to 3. In the description indicating an alloy, “-” is omitted in some cases. Specifically, it is described as “material1material2material3.”

(18) As illustrated in FIG. **1** and FIG. **2**, the light-emitting device **100** includes a substrate **10**, and the substrate **10** includes a plate-shaped first base material **13** and a second base material **14**, and an insulating first coating film **15**. The first base material **13** and the second base material **14** are made of copper (Cu) and arranged side by side to be mutually spaced. The insulating first coating film **15** covers them, and is a resin material made of a modified polyamide-imide. The first base material **13** and the second base material **14** are integrally covered with the insulating first coating film **15**. Specifically, the first coating film **15** is formed so as to cover upper surfaces and side surfaces of the first base material **13** and the second base material **14**. The first coating film **15** is filled in a clearance region between the first base material **13** and the second base material **14**, and mutually facing side surfaces of the first base material **13** and the second base material **14** are bound in the clearance region. That is, the first base material **13** and the second base material **14** are bonded together by a binding portion CB of the first coating film **15** filled in the clearance therebetween. The light-emitting device **100** includes a frame body **20** that is disposed in an outer peripheral portion on the upper surface of the substrate **10**, made of a material the same as that of the first base material **13** and the second base material **14**, and coated with a second coating film **24**.

(19) The light-emitting device **100** includes a light-emitting element **30** placed on an internal electrode **19** formed on the first base material, a wavelength converter **40** placed so as to cover a light emitting portion EM of the light-emitting element **30**, and a protection element **50** placed on an internal electrode (not illustrated) formed on the first base material.

(20) The first base material **13** and the second base material **14** are, for example, base materials of the substrate **10** having a rectangular planar shape in top view, and made of a metallic material such as copper (Cu). The first base material **13** and the second base material **14** are conductive bodies having a function of electrically connecting to a mounting board (not illustrated).

(21) As described above, the first base material **13** and the second base material **14** are coated with the first coating film **15** such that the upper surfaces and the side surface are covered. The first base material **13** and the second base material **14** are provided with the clearance between the mutually facing side surfaces, and the clearance is filled with the first coating film **15**. Since the first coating film **15** is an insulator, the first base material **13** and the second base material **14** are mutually electrically insulated while being mutually bound by the binding portion CB of the first coating film **15**.

(22) As illustrated in FIG. **2**, the first base material **13** and the second base material **14** include, for example, recessed structures HE provided by half etching in lower surface sides of the respective side surfaces. Accordingly, since the binding area of the binding portion CB to the first base material **13** and the second base material **14** are increased, the binding strength between the first base material **13** and the second base material **14** via the binding portion CB can be enhanced.

(23) The light-emitting device **100** includes the frame body **20** disposed in the outer peripheral portion of the upper surface of the substrate **10**, and the frame body **20** includes a frame base

material **23** and the second coating film **24**. The frame base material **23** is, for example, a metallic material the same material as the first base material **13** and the second base material **14**. The second coating film **24** is a resin material made of a modified polyamide-imide similarly to the first coating film **15**, and the frame base material **23** is coated with the second coating film **24** over the whole peripheral area of the frame base material **23**. That is, the substrate **10** is electrically insulated from the frame body **20**. For the substrate **10** and the frame body **20**, the first coating film **15** is bound to the second coating film **24** by pressurizing and heating. That is, the light-emitting device **100** is provided with a U-shaped cavity, which is open upward and formed by the substrate **10** and the frame body **20**, on the upper surface.

(24) While a case where the first base material **13**, the second base material **14**, and the frame base material **23** contain Cu as the main material is described in the embodiment, the material is not limited to this. For example, it is only necessary that the material is conductive and has high thermal conductivity, such as iron (Fe), aluminum (Al), iron-nickel (Fe—Ni) alloy, and iron-cobalt (Fe—Co) alloy. While a case where the first base material **13** and the second base material **14** are made of the same material as the frame base material **23** is described in the embodiment, the first base material **13** and the second base material **14** need not be made of the same material as the frame base material **23**.

(25) On the lower surfaces of the first base material **13** and the second base material **14**, external electrodes EL**1** and EL**2** are formed. The external electrodes EL**1** and EL**2** are electrodes made of, for example, nickel/gold (Ni/Au), and function as external electrodes electrically connected to a mounting board.

(26) The first coating film **15** is provided with an opening portion **16**A that exposes the region on which the light-emitting element **30** is placed on the upper surface of the first base material **13**. The opening portion **16**A is firmed in a shape approximately the same as the light-emitting element **30**, and formed so as to penetrate from the upper surface of the first coating film **15** to the upper surface of the first base material **13**. In the light-emitting element **30** placement region of the upper surface of the first base material **13** exposed from the opening portion **16**A, the internal electrode **19** as an element placing pad made of nickel/gold (Ni/Au) is formed. In the embodiment, a case where the light-emitting element **30** is placed in the center of the upper surface of the substrate **10**, and the protection element **50** is placed in the periphery of the light emitting element **30** will be described. The first coating film **15** is provided with an opening portion **16**B that exposes the upper surface of the first base material **13** in the region on which the protection element **50** is placed.

(27) The first coating film **15** on the upper surface of the second base material **14** is provided with an opening portion **16**C that exposes the upper surface of the second base material **14** in a region opposed to the light-emitting dement **30** across the binding portion CB. A bonding pad BP is disposed to the part exposed from the opening portion **16**C of the second base material **14**. The bonding pad BP is made of, for example, Ni/Au that is the material the same as that of the internal electrode **19**.

(28) The light-emitting element **30** is, for example, a semiconductor light-emitting element having the light emitting portion EM as a semiconductor light-emitting layer laminated on a support substrate **33** containing a conductive semiconductor as the main material. The light-emitting element **30** is bonded to the opening portion **16**A of the first coating film **15** via the internal electrode **19** made of nickel/gold (Ni/Au) as the element placing portion and an element bonding layer **60** as a conductive bonding member. The light emitting portion EM has, for example, a structure in which a p-type semiconductor layer, a light-emitting layer, and a n-type semiconductor layer are laminated. The upper surface of the n-type semiconductor layer is each of the upper surfaces of the light-emitting portions EM and functions as a light extraction surface in each of the light-emitting elements **30**. The p-type semiconductor layer, the light-emitting layer, and the n-type semiconductor layer are, for example, nitride semiconductors made of gallium nitride (GaN) or the like as the main material and blue color light-emitting diodes (LED) that radiate blue light from the

light-emitting layers having a multiple quantum well structure. The lower surface of the light-emitting element **30** is electrically connected to the p-type semiconductor via the support substrate **33**, and the lower surface of the light-emitting element **30** functions as a cathode electrode (not illustrated). That is, the cathode electrode being the lower surface of the light-emitting element **30** is electrically connected to the external electrode EL**1** via the element bonding layer **60** and the internal electrode **19**.

(29) The light-emitting element **30** includes an electrode pad **35** formed so as to be spaced from the light-emitting portion EM the support substrate **33**. The electrode pad **35** is electrically connected to the p-type semiconductor layer of the light-emitting portion EM and functions as an anode electrode of the light-emitting element **30**. The electrode pad **35** is, for example, electrically connected to the bonding pad BP formed on the second base material **14** via conductive bonding wires BW made of gold (Au). That is, the electrode pad **35** as a cathode electrode of the light-emitting element **30** is electrically connected to the external electrode EL**2** via the bonding wire BW and the bonding pad BP.

(30) As described above, the element bonding layer **60** as a bonding member is a conductive adhesive that fixedly secures and electrically connects the light-emitting element **30** to the internal electrode **19**. The element bonding layer **60** is, for example, a gold-tin alloy (AuSn alloy). Further, the element bonding layer **60** is, for example, a paste-like adhesive made of AuSn alloy particles and flux as the raw materials (hereinafter, the raw materials of the element bonding layer **60** may be simply referred to as an AuSn paste). In the embodiment, while a case where the element bonding layer **60** is made of the AuSn alloy is described, the element bonding layer **60** is not limited to this. For example, the element bonding layer **60** may be an argentum (Ag) paste or a solder paste. The element bonding layer **60** is a conductive adhesive having non-affinity to the first coating film **15**.

(31) The protection element **50** is for example, a reverse voltage protection element, such as a zener diode. When an overvoltage is applied to the light-emitting element **30** from outside (such as, static electricity), the protection element **50** operates so as to protect the light-emitting element **30**. The protection element **50** includes, for example, an electrode pad **51** that functions as a cathode electrode on the upper surface of the element, and an anode electrode (not illustrated) on the lower surface of the element. Further, similarly to the light-emitting element **30**, the anode electrode of the protection element **50** is bonded on the internal electrode (not illustrated) formed in the opening portion **16**B of the first coating film **15** via the element bonding layer **60** the same as one used for bonding the light-emitting element **30**, and the anode electrode of the protection element **50** is electrically connected to the external electrode EL**1**. The electrode pad **51** formed on the upper surface of the protection element **50** is electrically connected to the external electrode EL**2** via the bonding wire BW and the bonding pad BP. That is, the protection element **50** is connected in parallel and in a reversed polarity to the light-emitting element **30**.

(32) In the light-emitting element **30** and the protection element **50**, the bonding wire BW is configured by an aspect of what is called reverse bonding made of a wire bump and a gold wire. In the embodiment, while a case where the bonding wire BW is configured by reverse bonding is described, the aspect of the bonding wire BW is not limited to this and may be an aspect of forward bonding in which a press-bonded ball is formed on each electrode pad.

(33) The light-emitting device **100** includes a wavelength converter **40** placed on the light-emitting element **30** so as to cover the light emitting portion EM. The wavelength converter **40** performs wavelength conversion to emitted light from the light-emitting element **30**. The wavelength converter **40** is formed of a plate-shaped member containing, for example, phosphor particles containing yttrium aluminum garnet (YAG) doped with cerium (Ce) as the main material and a binder of glass or ceramic such as alumina that transmits radiation light of the light-emitting element **30** and radiation light of the phosphor particles. As the binder, YAG not doped with cerium (Ce) may be used (in this case, the wavelength converter **40** may be a polycrystal body or a single

crystal body). In the embodiment, the wavelength converter **40** is a wavelength conversion plate that performs the wavelength conversion of a part of blue light radiated from the light-emitting element **30**, and combines light radiated from the light-emitting element **30** and light radiated from the wavelength converter **40** to radiate white light. Performing the wavelength conversion of approximately all light radiated from the light-emitting element **30** by the wavelength converter **40** allows converting the light radiated from the wavelength converter **40** that is exposed to the outside of the light-emitting device **100** into green light, yellow light, red light, infrared light, and the like that correspond to the radiation light of the wavelength converter **40**.

(34) In the embodiment, the wavelength converter **40** has one main surface that is bonded on the upper surface of the light-emitting element **30** via an adhesive resin (not illustrated) as an adhesive resin layer that transmits the light radiated from the light-emitting element **30**. Further, the other main surface faces to the outside of the light-emitting device **100** so as to be exposed. That is, the one main surface of the wavelength converter **40** functions as a light-receiving surface that receives the light emitted by the light-emitting element **30** via the adhesive resin, and the other main surface functions as a light extraction surface of the light-emitting device **100**.

(35) As illustrated in FIG. **2**, the light-emitting device **100** includes a coating member **70** filled in the cavity formed by the substrate **10** and the frame body **20** so as to expose the upper surface as the light extraction surface of the wavelength converter **40**. The coating member **70** contains, for example, a resin material having reflectivity. In the embodiment, the coating member **70** is a white resin that reflects the light emitted from the light-emitting element **30** and the light emitted from the wavelength converter **40**. In FIG. **1**, the coating member **70** is omitted for clearly illustrating the structures and the positional relationship of the respective components.

(36) The light-emitting device **100** includes a thin film region **17**A in which the first coating film **15** is formed in a thin film in an outer peripheral portion of the opening portion **16**A of the first coating film **15**. The thin film region **17**A has a function of easily performing self-alignment when the light-emitting element **30** is placed on the paste-like element bonding layer **60** applied over the upper surface of the internal electrode **19** and the element bonding layer **60** is heated to be melted.

(37) The opening portion **16**A of the first coating film **15** is opened in a shape approximately the same shape as the element to be placed, such as the light-emitting element **30**. Assuming that the thin film region **17**A is not formed around the opening portion **16**A of the first coating film **15**, an interference between the light-emitting element **30** and the first coating film **15** occurs or the light-emitting element **30** runs over the end portion of the coating film due to the positional deviation or the like in placing the element, thereby possibly causing a poor bonding.

(38) Therefore, forming the thin film region **17**A on the outer periphery of the opening portion **16**A of the first coating film **15** and using the element bonding layer **60** (AuSn paste) having the non-affinity to the first coating film **15** allow the element to be placed to easily perform the self-alignment. Accordingly, the light-emitting element **30** can be stably placed on the substrate **10** without causing the positional deviation of the light-emitting element **30** placed on the substrate. Also in the opening portion **16**B on which the protection element **50** is placed, a thin film region **17**B is similarly formed.

(39) As described above, the light-emitting device **100** includes the plate-shaped first base material **13** and second base material **14**, the insulating first coating film **15**, the internal electrode **19**, and the light-emitting element **30**. The first base material **13** and second base material **14** are arranged side by side to be mutually spaced, and are each made of a metal. The first coating film **15** is formed so as to cover the upper surface and the side surface of each of the first base material **13** and the second base material **14**. The first coating film **15** is provided with the opening portion **16**A that exposes one region of the upper surface of the first base material **13** among the first base material **13** and the second base material **14**. The first coating film **15** includes the binding portion CB binding the first base material **13** to the second base material **14**. The internal electrode **19** is disposed so as to cover the one region of the upper surface of the first base material **13** and made of

a metal. The light-emitting element **30** is placed on the internal electrode **19** via the element bonding layer **60**. The first coating film **15** includes the thin film region **17**A in which the first coating film **15** is formed in a thin film so as to surround the outer peripheral end of the opening portion **16**A.
(40) The first coating film **15** is made of a modified polyamide-imide. The first coating film **15** has a non-affinity to the element bonding layer **60** one another.
(41) The height from the upper surface of the first base material **13** to the bottom surface of the light-emitting element **30** is higher than the height from the upper surface of the first base material **13** to the upper surface of the thin film region **17**A. The light-emitting device **100** further includes the frame body **20** disposed on the outer peripheral portion of the upper surface of the first coating film **15**. The frame body **20** includes the annular frame base material **23** coated with the second coating film **24**. The frame base material **23** is made of the material the same as those of the first base material **13** and the second base material **14**. The opening portion **16**A has a shape approximately the same as the shape of the light-emitting element **30**.
(42) Here, exemplary configurations of the respective layers of the first coating film **15**, the thin film region **17**A, the internal electrode **19**, and the element bonding layer **60** in the light-emitting device **100** will be described.
(43) FIG. **3** is an enlarged view of a peripheral portion PA of an element placing portion of FIG. **2**.
(44) As described above, the first coating film **15** is formed on the first base material **13**, and the first coaling film **15** is provided with the opening portion **16**A having approximately the same shape as the support substrate **33** of the light-emitting element **30** and opening so as to penetrate to the upper surface of the first base material **13**. The internal electrode **19** made of Ni/Au is formed in the opening portion **16**A of the first coating film **15**. The support substrate **33** of the light-emitting element **30** is bonded on the internal electrode **19** via the element bonding layer **60**. The thin film region **17**A is formed at the outer peripheral portion of the opening portion **16**A of the first coating film **15**.
(45) Here, a film thickness of the thin film region **17**A from the upper surface of the first base material **13** is formed to be lower than a thickness of the element bonding layer **60** including the internal electrode **19** from the upper surface of the first base material **13**.
(46) Since the element bonding layer **60** has the non-affinity to the first coating film **15**, the element bonding layer **60** does not wet-spread to the thin film region **17**A outside the internal electrode **19**. In other words, the element bonding layer **60** wet-spreads only over the internal electrode **19** even during the heating and melting, and has the shape similar to the opening portion **16**A of the first coating film **15** (upper surface shape of the light-emitting element **30**) in top view.
(47) A height from the upper surface of the first base material **13** to the upper surface of the first coating film **15** is configured to be higher than a height including the internal electrode **19** and the element bonding layer **60** from the upper surface of the first base material **13** (height from the upper surface of the first base material **13** to the lower surface of the support substrate **33** of the light-emitting element **30**).
(48) Accordingly, it can be avoided that the light-emitting element **30** deviates from the inside of the opening including the thin film region **17**A to the outside (upper surface of the first coating film **15**). For example, the displacement of the light-emitting element **30** due to a vibration or the like can be avoided after the light-emitting element **30** is placed on the element bonding layer **60** by a placing machine.
(49) Accordingly, the light-emitting element **30** is self-aligned by a surface tension having a minimum interfacial energy of the melted element bonding layer **60** between the heated and melted element bonding layer **60** and the light-emitting element **30** placed on the element bonding layer **60**.
(50) In the region where the protection element **50** is placed, the first coating film **15**, the thin film region **17**B, the internal electrode, and the element bonding layer **60** have the similar

configurations.
(51) While the case where the film thickness of the thin film region **17**A from the upper surface of the first base material **13** is smaller than the film thickness of the internal electrode **19** is illustrated in the embodiment, the film thickness of the thin film region **17**A may be thicker than the film thickness of the internal electrode **19**. Specifically, it is only necessary that the height of the bottom surface of the light-emitting element **30** is higher than the height of the thin film region **17**A when the AuSn paste is melt. In other words, it is only necessary to have a configuration in which the light-emitting element **30** does not contact the modified polyamide-imide of the thin film region **17**A in the self-alignment of the light-emitting element **30**.
(52) In the embodiment, the case where the opening portion **16**A has approximately the same shape as the support substrate **33** of the light-emitting element **30** will be described. However, the shape of the opening portion **16**A may be approximately the same shape as a cathode electrode (not illustrated) on the lower surface of the light-emitting element **30**. In this case also, since the height of the bottom surface of cathode electrode is higher than the height of the thin film region **17**A when the AuSn paste is melted, the accurate self-alignment can be performed similarly to the embodiment.
(53) A width in which the thin film region **17**A is formed is, for example, a width providing an overlapping area that is 60% or more of an area defined by the outer shape of the light-emitting element **30** relative to the surface on which the element bonding layer **60** is formed in top view.
(54) For example, when the outer shape of the light-emitting element **30** is a square of 1 mm×1 mm, and the outer shape of the element bonding layer **60** is a square of 1 mm×1 mm, it is only necessary to cause the width of the thin film region **17**A to have 0.2 mm or less for obtaining the overlapping area of 64% or more. Thus, setting the overlapping area to 60% or more allows the self-alignment of the light-emitting element **30** on the element bonding layer **60**.
(55) In the embodiment, since the outer shape of the light-emitting element **30** is 1 mm×1 mm, the outer shape of the element bonding layer **60** is 1 mm×1 mm, and the width of the thin film region **17**A is 0.04 mm, the overlapping area is 92%. Thus, setting the overlapping area to more than 90% allows the extremely accurate self-alignment. When the width of the thin film region **17**A is narrowed, fan-shaped relief portions centered at intersection points of four sides defining the thin film region **17**A may be provided. With the relief portions, it can be avoided that the corner portion of the light-emitting element **30** runs over the upper surface of the first coating film **15**.
(56) Next, a manufacturing procedure of the light-emitting device **100** according to the embodiment of this application will be described with reference to FIG. **4** and FIG. **5** to FIG. **15**.
(57) FIG. **4** is a drawing illustrating a flow chart of manufacturing the light-emitting device **100** according to the embodiment of the present invention. FIG. **5** to FIG. **15** illustrate top views of the light-emitting device **100** in respective steps of the manufacturing procedure illustrated in FIG. **4**.
(58) First, the plate-shaped first base material **13** and second base material **14** that are arranged side by side to be mutually spaced and made of a metal are prepared as illustrated in FIG. **5** (Step S**101**).
(59) Next, as illustrated in FIG. **6**, the insulating modified polyamide-imide is applied over the side surfaces and the upper surfaces of the first base material **13** and the second base material **14**, thus forming the first coating film **15** and the binding portion CB (Step S**102**). The modified polyamide-imide is formed using electrodeposition coating. The film thickness of the modified poly amide-imide formed by the electrodeposition coating is, for example, about 20 um to 40 um. The lower surface sides of the first and second base materials **13** and **14** not to be coated with the first coating film **15** are, for example, masked in advance.
(60) As illustrated in FIG. **7**, with the method similar to the first base material **13** and the second base material **14**, the frame body **20** in which the second coating film is coated over the frame base material **23** is prepared. Then, after positioning along the outer peripheral portion of the upper surface of the substrate **10**, heating and pressurizing are performed to both, thereby pressure-bonding the frame body **20** to the substrate **10** (Step S**103**).

(61) Next, as illustrated in FIG. **8**, a laser beam LB is irradiated on a first region including the region in which the light-emitting element **30** is placed and the region of its outer periphery corresponding to the thin film region **17**A on the upper surface of the first base material **13**, and the region in which the bonding pad BP is formed on the upper surface of the second base material **14**, thereby removing the first coating film **15** (Step S**104**). The laser beam LB for example, a laser light of Yttrium Aluminum Garnet (YAG). At this time, an output of the laser beam LB is set so as to partially leave the first coating film **15**. The output is set such that the film thickness of the remaining first coating film **15** becomes the film thickness of the thin film region **17**A. For example, in the removal by the laser beam LB, since the removal rate is decreased due to the heat radiation from the first and second base materials **13** and **14** as the first coating film **15** becomes thin, the setting to leave the thin film region **17**A can be easily made. While the method for removing the first coating film **15** using the YAG laser as the laser beam LB is described in the embodiment, the type of the irradiated laser beam is not limited to this. Specifically, it is only necessary that the modified polyimide-imide as the first coating film **15** can be removed by the irradiation of the laser beam LB.
(62) Next, as illustrated in FIG. **9**, the laser beam LB is irradiated again on a second region including the region in which the light-emitting element **30** is placed on the upper surface of the first base material **13**, and the region in which the bonding pad BP is formed on the upper surface of the second base material **14**, thereby completely removing the first coating film **15** (Step S**105**). That is, the first coating film **15** is removed so as to expose the upper surfaces of the first base material **13** and the second base material **14**. Accordingly, the opening portion **16**A opened so as to penetrate to the first base material **13** and the thin region **17**A in a thin film shape at the outer peripheral portion of the opening portion **16**A can be formed on the first coating film **15** formed on the upper surface of the first base material **13**. As described above, the upper surface shape of the opening portion **16**A formed on the first coating film **15** is formed in approximately the same shape as the upper surface shape of the light-emitting element **30** to be placed. Similarly, in the region where the bonding pad BP is formed on the second base material **14**, the opening portion **16**C opened so as to penetrate to the upper surface of the second base material **14** is formed.
(63) While the case where the opening portion **16**A and the thin film region **17**A are formed on the first coating film **15** by performing the laser processing step twice as described above is described in the embodiment, the number of times of performing the laser processing is not limited to this. Specifically, the process may be performed by the laser irradiation at one time while changing the output of the laser beam LB for the opening portion **16**A and the thin film region **17**A as needed, or the process may be performed by a plurality of times of laser irradiation, three times or more, with a predetermined laser output. In the embodiment, the case where the first coating film **15** is formed only on the upper surfaces and the side surfaces of the first base material **13** and the second base material **14** is described. However, the range of forming the first coating film **15** is not limited to this. For example, it is allowed that the first coating film **15** is formed on the whole circumferences including the lower surfaces of the first base material **13** and the second base material **14**, and then, the first coating film **15** is removed by the laser irradiation in the region in which the external electrodes EL**1** and EL**2** are formed on the lower surfaces of the first base material **13** and the second base material **14**.
(64) Steps S**101** to S**105** described above are performed as a substrate forming step of forming the substrate **10**.
(65) Next, as a plating step, as illustrated is FIG. **10**, Ni/Au plating is performed on the substrate **10** on which the laser processing has been performed, thus forming the internal electrode **19** in the opening portion **16**A on the upper surface of the first base material **13**, the bonding pad BP in the opening portion **16**C of the second base material **14**, and the external electrodes EL**1** and EL**2** on the lower surfaces of the first base material **13** and the second base material **14** (Step S**106**).
(66) Next, as illustrated in FIG. **11**, the AuSn paste as the raw material of the element bonding layer

60 is applied over the internal electrode **19** formed on the upper surface of the first base material **13** in the opening portion **16**A (Step S**107**). When the AuSn paste is applied over the internal electrode **19**, applying using a dispenser filled with the AuSn paste is preferred.
(67) Next, as illustrated in FIG. **12**, the light-emitting element **30** is placed on the element bonding layer **60** applied over the internal electrode **19** such that the cathode electrode (not illustrated) disposed on the lower surface of the light-emitting element **30** contacts the element bonding layer **60** (Step S**108**).
(68) The substrate **10** in this state is, for example, heated to 300° C. in a nitrogen atmosphere to melt AuSn alloy particles in the AuSn paste, and then cooled, thereby forming the element bonding layer **60**. Thus, the light-emitting element **30** is fixedly secured to the internal electrode **19** by the element bonding layer **60**. The internal electrode **19** and the light-emitting element **30** are eutectically bonded is one another by the AuSn alloy, and electrically connected to one another. The position and the orientation of the secured light-emitting element **30** are, as described above, those allowing the self-alignment of the light-emitting element **30** by the surface tension of the melted element bonding layer **60**. Specifically, the light-emitting element **30** is self-aligned by a surface tension having a minimum interfacial energy of the melted element bonding layer **60** between the heated and melted element bonding layer **60** and the light-emitting, element **30** placed on the element bonding layer **60**. Additionally, the p-type semiconductor layer of the light emitting portion EM is electrically connected to the external electrode EL**1**.
(69) Since the modified polyamide-imide used for the first coating film **15** and the binding portion CB has a heat resistance at a temperature higher than the temperature of the eutectic bonding of AuSn alloy, the failure such as degeneration is not caused even in the formation of the element bonding layer **60**.
(70) Steps S**107** to S**108** described above are performed as an element bonding step of bonding the light-emitting element **30** to the substrate **10**.
(71) Next, as illustrated in FIG. **13**, the substrate **10** to which the light-emitting element **30** is fixedly secured is set in a wire bonding machine, and the electrode pad **35** formed on the upper surface of the light-emitting element **30** is connected to the bonding pad BP formed on the second base material **14** with the bonding wire BW, such as an Au wire (Step S**109**). The connection by the Au wire may be the aspect of forward bonding in which a press-bonded ball is formed on the electrode pad **35**, or may be the aspect of reverse bonding in which, after a wire bump is formed on the electrode pad **35**, the press-bonded ball is formed on the bonding pad BP and connected to the wire bump. Thus, the n-type semiconductor layer of the light emitting portion EM is electrically connected to the external electrode EL**2**.
(72) Next, as illustrated in FIG. **14**, the wavelength converter **40** is placed so as to cover the light emitting portion EM of the light-emitting element **30** via a not illustrated adhesive resin, and then, heated to be fixedly secured (Step S**110**). The adhesive resin is preferably potted on, for example, the center position or the like of the light-emitting portion EM by a dispenser. After the potting of the adhesive resin, the wavelength converter **40** is placed on the adhesive resin. The placing position of the wavelength converter **40** is self-aligned so as to cover the upper surface of the light-emitting element **30**, especially the upper surface of the light emitting portion EM by the surface tension of the adhesive resin. Afterwards, the substrate **10** in which the wavelength converter **40** is placed on the upper surface of the light-emitting element **30** is heated to cure the adhesive resin, and the wavelength converter **40** is fixedly secured on the upper surface of the light-emitting element **30**.
(73) Subsequently, as illustrated in FIG. **15**, the reflective coating member **70** is filled inside the cavity formed by the substrate **10** and the frame body **20**, thus manufacturing the light-emitting device **100** (Step S**111**).
(74) Although not illustrated, the processes of Steps S**104** to S**109** are performed also for the protection element **50**, and the protection element **50** is bonded on the first base material **13** so as to

be self-aligned by the thin film region **17**B formed in the outer periphery of the opening portion **16**B in the region where the protection element **50** is placed.

(75) As described above, the method for manufacturing the light emitting device **100** includes: a substrate forming step of forming the plate-shaped first base material **13** and second base material **14** that are arranged side by side to be mutually spaced and are each made of a metal, and forming the insulating first coating film **15** formed so as to cover the upper surface and the side surface of each of the first base material **13** and the second base material **14**, the first coating film **15** being provided with the opening portion **16**A that exposes one region of the upper surface of the first base material **13** among the first base material **13** and the second base material **14**, the first coating film **15** including the binding portion CB binding the first base material **13** to the second base material **14**; a plating step of providing the internal electrode **19** made of a metal so as to cover the one region of the upper surface of the first base material **13**; and an element bonding step of placing the light-emitting element **30** on the internal electrode **19** via the element bonding layer **60**. In the substrate forming step, the thin film region **17**A in which the first coating film **15** is formed in a thin film so as to surround the outer peripheral end of the opening portion **16**A is formed in the first coating film **15**.

(76) The first coating film **15** and the binding portion CB are formed on the first base material **13** and the second base material **14** by electrodeposition coating.

(77) The thin film region **17**A and the opening portion **16**A of the first coating film **15** are formed by removing the first coating film **15** by the irradiation of the laser beam LB to the first coating film **15**, the thin film region **17**A is formed by irradiating a laser irradiation region including the one region of the first coating film **15** and extending outward from the peripheral area of the one region with the laser beam LB, and the opening portion **16**A is formed by irradiating the one region in a laser irradiation region with the laser beam LB again to partially remove the thin film region **17**A.

(78) According to the embodiment, in the light-emitting device **100**, in the region, in which the light-emitting element **30** is placed, of the insulating first coating film **15** coated over the upper surface of the first base material **13**, the first coating film **15** is provided with the opening portion **16**A having approximately the same shape as the light-emitting element **30** to be placed and penetrating to the upper surface of the first base material **13**. The thin film region **17**A provided by forming the first coating film **15** in a thin film is disposed on the upper surface of the first base material **13** in the outer peripheral portion of the opening portion **16**A. The film thickness of the thin film region **17**A is smaller than the thickness from the upper surface of the first base material **13** to the bottom surface of the light-emitting element **30**. With this configuration, even when the placing position is displaced in placing the light-emitting element **30**, the poor bonding due to the contact of the light-emitting element **30** with the first coating film **15** or the like can be avoided, thus allowing surely performing the self-alignment during heating.

(79) Accordingly, the present invention can provide the light-emitting device **100** capable of stably placing the light-emitting element **30** on the substrate **10** without causing the positional deviation of the light-emitting element **30** placed on the substrate **10**, and provide the method for manufacturing the light-emitting device **100**.

(80) While the case where the light emitting portion EM of the light-emitting element **30** is a blue LED containing a nitride semiconductor as a main material is described in the embodiment, the material of the light emitting portion EM is not limited to this, and various kinds of LED emitting lights of other colors and a semiconductor light-emitting layer of laser are applicable. Specifically, it is only necessary for the light-emitting element **30** that the light emitting portion EM and the electrode pad **35** are placed side by side on the support substrate **33**.

(81) In the embodiment, the light-emitting device **100** including the wavelength converter **40** configured to perform the wavelength conversion of a light radiated from the light-emitting element **30** upon excitation by the radiated light is described. However, the wavelength converter

40 may be a translucent plate that does not perform the wavelength conversion of the light radiated from the light-emitting element **30**.
(82) In the embodiment, the case where the light-emitting device **100** includes one light-emitting element **30** is described. However, the number of the light-emitting elements **30** to be mounted is not limited to this. Specifically, a plurality of two or more light-emitting elements may be mounted on the first base material **13**. In this case, the plurality of light-emitting elements are configured to be connected in parallel to the external electrode.
(83) In addition to the above-described configuration, for the base material constituting the substrate **10**, three or more base materials may be included. In this case, the light-emitting device **100** may have a configuration in which a base material including an external electrode on a lower surface thereof and a base material without an external electrode on a lower surface thereof are prepared, and a plurality of light-emitting elements are connected in series to the respective base materials. The light-emitting device may include a plurality of light-emitting elements connected in parallel and each including an external electrodes on a lower surface thereof.
(84) When the light-emitting device includes a plurality of light-emitting elements, as the wavelength converter **40**, a wavelength converter may be disposed to each of the plurality of light-milting elements. Wavelength converters integrally formed so as to cover the respective light emitting portions EM of a plurality of light-emitting elements may be disposed while the plurality of light-emitting elements are placed side by side in a specific arranging direction.
(85) A plurality of light-emitting elements may be placed side by side in a specific arranging direction, and wavelength converters may be disposed for of the respective plurality of light-emitting elements. In this case, wavelength conversion plates excited by lights radiated from the light-emitting elements to perform the wavelength conversion into white lights and wavelength conversion plates excited by lights radiated from the light-emitting elements **30** to perform the wavelength conversion into orange lights may be alternately disposed for the plurality of respective light-emitting elements. This allows toning the light emitted from the light extraction surface of the light-emitting device **100**.

Claims

1. A light-emitting device comprising: a plurality of plate-shaped base materials that are arranged side by side to be mutually spaced and are each made of a metal; an insulating coating film formed so as to cover an upper surface and a side surface of each of the plurality of base materials, the coating film being provided with an opening portion that exposes one region of the upper surface of one base material among the plurality of base materials, the coating film including a binding portion mutually binding the plurality of base materials; a placing pad disposed so as to cover the one region of the upper surface of the one base material, the placing pad being made of a metal; and at least one light-emitting element placed on the placing pad via a bonding member, wherein the coating film includes a thin film portion in which the coating film is formed in a thin film so as to surround an outer peripheral end of the opening portion.
2. The light-emitting device according to claim 1, wherein the coating film is made of a modified polyamide-imide.
3. The light-emitting device according to claim 1, wherein the coating film has a non-affinity to the bonding member.
4. The light-emitting device according to claim 1, wherein a height from the upper surfaces of the plurality of base materials to a bottom surface of the at least one light-emitting element is higher than a height from the upper surfaces of the plurality of base materials to an upper surface of the thin film portion.
5. The light-emitting device according to claim 1, further comprising: a frame body disposed on an outer peripheral portion of the upper surface of the coating film, the frame body including an

annular frame base material coated with the coating film, the frame base material being made of a material same as the material of the plurality of base materials, wherein the opening portion has a shape approximately same as a shape of the at least one light-emitting element.

6. A method for manufacturing a light-emitting device, comprising: a substrate forming step of forming a plurality of plate-shaped base materials that are arranged side by side to be mutually spaced and are each made of a metal, and forming an insulating coating film formed so as to cover an upper surface and a side surface of each of the plurality of base materials, the coating film being provided with an opening portion that exposes one region of the upper surface of one base material among the plurality of base materials, the coating film including a binding portion mutually binding the plurality of base materials; a plating step of providing a placing pad made of a metal so as to cover the one region of the upper surface of the one base material; and an element bonding step of placing at least one light-emitting element on the placing pad via a bonding member, wherein in the substrate forming step, a thin film portion in which the coating film is formed in a thin film so as to surround an outer peripheral end of the opening portion is formed in the coating film.

7. The method for manufacturing a light-emitting device according to claim 6, wherein the coating film and the binding portion are formed on the plurality of base materials by electrodeposition coating.

8. The method for manufacturing a light-emitting device according to claim 6, wherein: the thin film portion and the opening portion of the coating film are formed by removing the coating film by the irradiation of a laser beam to the coating film, and the thin film portion is formed by irradiating a laser irradiation region including the one region of the coating film and extending outward from a peripheral area of the one region with the laser beam, and the opening portion is formed by irradiating the one region in the laser irradiation region with the laser beam again to partially remove the thin film portion.